

This is to build up some intuition about option pricing and trading strategies. We will use the following notation:

- S_0 : Current stock price
- K : Strike price of option
- T : Time to expiration of option
- S_T : Stock price at maturity
- r : Continuously compounded risk-free rate of interest for an investment maturing in time T
- C : Value of American call option to buy one share
- P : Value of American put option to sell one share
- c : Value of European call option to buy one share
- p : Value of European put option to sell one share

1 Basic Boundaries

- American options are more valuable than European options: $C \geq c, P \geq p$
- call options can't be worth more than the stock: $C \leq S_0, c \leq S_0$
- put options gives the right to sell one share for K , so it can't be worth more than K : $p \leq K, P \leq K$

2 Put-Call parity

Consider the following 4 portfolios for a non-dividend-paying stock

- Portfolio A: one European call option plus a amount of cash equal to Ke^{-rT}
- Portfolio B: one European put option plus one share
- Portfolio C: one share
- Portfolio D: an amount of cash equal to Ke^{-rt}

These 4 portfolios should be very reasonable and intuitive to think about. Portfolio A holds a call with potential cash to buy stock. Portfolio B holds a put with a share to potentially sell. Portfolio C holds a single share while portfolio D holds just cash. At time T , these 4 portfolios are:

- Portfolio A: if $S_T \leq K$, option worths nothing, the cash equals to K . if $S_T > K$, we exercise the option by using the cash that is equal to K to to buy one share which is valued at S_T . So the value of the portfolio is $\max(S_T, K)$

- Portfolio B: if $S_T \geq K$, option worths nothing, we have one share that is valued at S_T . if $S_T < K$, we exercise the option by selling the share at K and we hold cash equal to K at the end. So the value of the portfolio is $\max(S_T, K)$.
- Portfolio C: one share
- Portfolio D: an amount of cashe equal to K

2.1 Boundaries for European options